DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

NITK-SURATHKAL

II SEM M.Tech (CSE-IS) Mid-Semester Examinations

Sub: Network Security — Max Marks : 100

Course Code : CS851 Date:24-02-2026 — Duration : 1.5 hrs

Roll No: 252IS032

Note: Answer all the questions. Missing data may be suitably assumed.

1. Compare **ECB** with **CBC and CTR modes** in terms of:
   Confidentiality, Error propagation, Parallelization, and Security against pattern leakage

   10 Marks

2. A web application encrypts session tokens deterministically.
   Explain how a Chosen Plaintext Attack adversary can distinguish between two chosen inputs.

   --15 marks

3. Consider RSA without padding:
   $C=M^e \bmod n$
   Show how RSA is vulnerable under chosen ciphertext attack. –15 Marks

4. The OFB encryption process is defined as:
   $O_1=E_K(IV)$
   $O_i=E_K(O_{i-1})$
   $C_i=P_i \oplus O_i$
   (a) Show that encryption and decryption use the same keystream.
   (b) Explain why OFB does not propagate errors. --20 Marks

5. Show that CFB is vulnerable to chosen ciphertext attacks if authentication is absent.

   –10Marks

6. Explain why CFB without MAC is vulnerable to bit-flipping attacks. --10 Marks

7. A military communication system uses AES/DES-CFB mode for secure radio transmission.
   (a) Analyze its confidentiality guarantees.
   (b) Evaluate error propagation characteristics.
   (c) Assess vulnerability under chosen ciphertext attack.
   (d) Propose a secure redesign using authenticated encryption.

   --20 Marks

1. Peggy wants to prove to Victor that she knows an 'x' that satisfies $A^x \equiv B \bmod p$, where p is prime and x is a random number relatively prime to $p-1$. The numbers A, B, p are public, and x is secret. Give the zero knowledge proof — 10

2. Give the zero knowledge proof of Integer Factorization Problem. — 10.